

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Examination, 2016
CE 2107
(Construction Technology and Materials)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any **three** questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

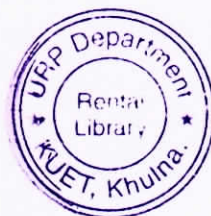
Section – A

1. (a) What are meant by "Natural Bed of Stones" and "Dressing of Stones"? Why the study of "Preservation of Stones" is important? Write down the important properties of an ideal preservative. (10)
(b) Write down the comparison between Kiln burning and clamp burning. Describe the working principles of Fixed Chimney Bull's Trench Kiln with neat sketch. (12)
(c) Discuss the relative advantages and disadvantages of bricks as compared with stones as an engineering material. (06)
(d) What is meant by "Defects in timber"? Describe as brief. (07)
2. (a) Define the terms flowability, compactability, pumpability and workability of fresh concrete and hack drying. (05)
(b) What is meant by "Fresh concrete"? What are the important properties to be considered in fresh concrete? (06)
(c) Describe six important factors which affect the workability of concrete. (12)
(d) Write down the recommended values of slump for various types of construction as given by ACI 211.1-81 (06)
(e) Write short note on "Limitations of concrete". (06)
3. (a) What is curing of concrete? Give a comparison between cube specimen and a cylinder specimen in concrete compressive strength test. (10)
(b) What are the important points to be considered while making selection of a suitable site for a cement factory? (07)
(c) What are the functions of sand in mortar? Why is it not wise to use sea sand for the preparation of concrete or mortar? (07)
(d) What is bulking of sand? Draw the qualitative gradation curve for well graded, gap graded and uniformly graded sample. (06)
(e) What is the significance of "Texture Triangle"? (05)
4. (a) Why uniaxial tension test of concrete is more difficult? How can you improve the tensile strength of concrete? (05)
(b) Classify the concrete in accordance with compressive strength and use. (07)
(c) What is the difference between "wood" and "timber"? State the characteristics of good timber. (06)
(d) What is meant by "Pedology"? Write down the advantages and shortcomings of plastics. (10)
(e) Choose a Ferrous and Non-Ferrous metal from the following table. Describe the suitable use of each metal with reasons. (07)

Ferrous Metal	Wrought Iron	Mild steel	Stainless Steel
Non-Ferrous Metals	Aluminum	Copper	Brass

Section – B

5. (a) What is meant by the term “foundation”? What are the purposes of foundation? (08)
- (b) When do you suggest trapezoidal shape of combined footing in lieu of rectangular shape? (06)
- (c) Differentiate with neat sketches: (07)
- i. Shallow and deep foundation
 - ii. Precast and cast in-situ pile
- (d) What is meant by “safe bearing capacity” of a soil? Describe plate load test for determining the safe bearing capacity of soil at site. (14)
6. (a) What are the materials that are used for masonry construction? How will you ensure the quality of those materials? What are the steps to follow for good masonry construction? (15)
- (b) Define with neat sketches: (08)
- i. Bevelled closer
 - iii. Coping
 - ii. Queen closer
 - iv. Gable
- (c) Write short notes on (with neat sketches): (12)
- i. Flemish bond
 - ii. Raking bond
 - iii. Herring bone bond
7. (a) What are the functions of lintels and arches? (05)
- (b) Distinguish between: (06)
- i. Intrados and Extrados
 - ii. Voussoirs and Keystone
- (c) Write short notes on: (09)
- i. Flat arch
 - ii. Four centered arch
 - iii. Concrete arch
- (d) Mention with sketches the different types of RCC floor system. (08)
- (e) Mention the most common types of trenching hazards and trenching collapse. (07)
8. (a) What are the merits and demerits of tiles flooring? Where do you suggest the location of doors in a building? (10)
- (b) Write short notes on: (12)
- i. Plaster of Paris
 - ii. Efflorescence
 - iii. Cantilever bridge
 - iv. Suspension bridge
- (c) What are the objectives of plastering? Describe the construction procedure of cement plaster and lime plaster. (13)
-



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2016
HUM 2117
(Sociology for Planners)

Full Marks. 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define Sociology. Discuss the nature and scope of Sociology. (15)
(b) Define Society. Explain different types of Society with their character. (20)
2. (a) What do you mean by association and institution? Find out the differentiation between association and institution. (15)
(b) What is family? What are the duties and responsibilities of a family? (10)
(c) What are the vital role of community and association for a family? (10)
3. (a) What is social structure? Explain the elements of social structure. (10)
(b) What is marriage? Why marriage is important in our society? Explain. (10)
(c) Discuss the merits and demerits of joint family. Do you think that the joint family system is changing? Why? (15)
4. (a) What is social stratification? Why are human society stratified? (10)
(b) What are the purposes of social control? What are the roles of agencies to create social control in our societies? (15)
(c) Discuss the different types of marriage. Which of them do you like most? Why? (10)

Section – B

5. (a) Define culture. (05)
(b) What is cultural relativism? (10)
(c) Explain the elements of culture. (20)
6. (a) What is Industrialization? (05)
(b) Explain compositional model and subcultural view about urbanism. (15)
(c) Explain Louis Wirth's view about urbanism. (15)
7. (a) What is Rural-Urban continuum? (10)
(b) What is suburban living and what are the problems of suburban living? (10)
(c) Explain the empirical consequences of urban living. (15)
8. (a) What is Invasion and succession process of a city? (10)
(b) Explain spatial distribution of a city. (10)
(c) What is "Modernization"? What are the impacts of modernization in a social structure? (15)



Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 2nd Year 1st Term Regular Examination, 2016

URP 2113
(Site and Area Planning)

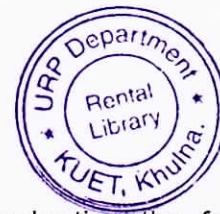
Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Explain the four basic models of site planning with the help of proper diagrams. (10)
(b) Give an outline of the site planning process as described by John Simonds. (05)
(c) Outline the benefits of Site selection. How can these benefits weigh in to facilitate the process of site selection? (15)
(d) Find the void ratio with the following data: $V_A=5m^3$, $V_W=3000000cm^3$ and $V_S=10m^3$. (05)
2. (a) Explain the various structural and drainage characteristics that a site planner must be aware of. (10)
(b) "Open space has many values and functions"- Explain this statement with an appropriate diagram. (10)
(c) How will you follow the guiding principles of recreation, parks, and open space (RPOS) for developing a park in Khulna City? (15)
3. (a) With the help of appropriate diagrams explain Burgess Zone Model and Hoyt Sector Model. (14)
(b) What are the key RPOS planning challenges? (06)
(c) "Ecological sustainability can be achieved through land use controls" – Justify the statement. (05)
(d) "State land use controls are passive land use controls"- Give your answer for or against this statement. (10)
4. (a) Describe site selection process with a flow diagram. (10)
(b) The Sixty Dome Mosque is a UNESCO World Heritage Site. Write something about its
 - Spirit
 - Character
 - Nature of the project
 - Behavioral studies(15)
(c) Prepare a counter plan for air movement by evaluating the following parameters for wind rose diagram: (10)
N -5,
NNE – 10,
NE – 15,
ENE – 50,
E – 10 (unit – m/hr)



Section – B

5. Write down the objectives, performance criteria and prescriptive measures with proper figure and explanation for **any five** of the following elements of residential subdivision planning. (35)
- (a) Allotment design
 - (b) Setbacks
 - (c) Solar Access
 - (d) Land Sensitivity
 - (e) Drainage Control
 - (f) Urban Streams
 - (g) Soil and Water Management
6. (a) "Site planning should be carried out in connection with the nature" - Narrate the statement with your rationalities. (10)
- (b) Which pattern of subdivision plan is more suitable than others in case of Bangladesh especially in the flat region as per your understanding and why? State your rationalities. (10)
- (c) Annotate any three of the following pattern of subdivision. (15)
- (i) Loop street pattern
 - (ii) Cluster pattern
 - (iii) Gridiron Pattern
 - (iv) Modified grid pattern
 - (v) Curvilinear pattern
7. (a) What are the regulatory bases of Private Residential Land Development Rules 2012? (05)
- (b) "Neighborhood is a nucleus for the development of the local social life of the environment" – Explain the statement with relations to site and area planning. (10)
- (c) Write down the associated maps, designs and plans other than detailed layout plan while applying for the approval of private residential housing project. (20)
8. (a) Write down the general conditions for the approval of a Private Residential Housing Project under the "Private Residential Housing Land Development Rules 2012". (15)
- (b) Chalk out three comparative critical examples (good design and bad design) of subdivision planning and neighborhood design with proper figure and explanation. (10)
- (c) Briefly explain the model of Site Planning Process by drawing proper diagram to show the relationship among the different process components. (10)



Khulna University of Engineering & Technology
Department of Urban & Regional Planning
BURP 3rd Year 1st Term Examination, 2016.
URP 2115
(Urban Planning Techniques)

Full Marks: 210

Time: 3 hrs

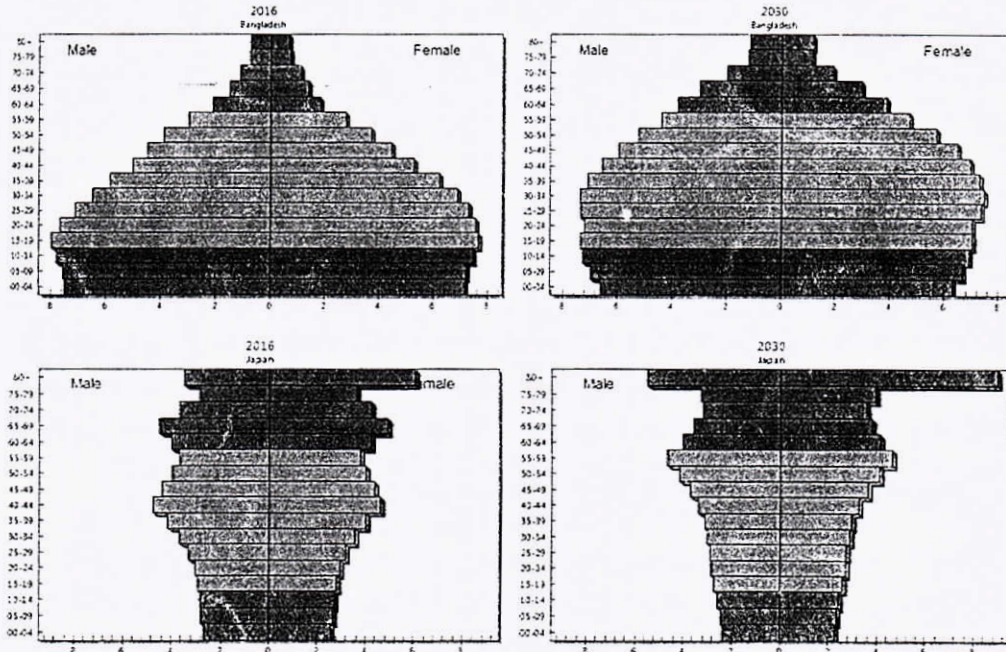
- N.B.** i) Figures in the right margin indicate full marks.
ii) Answer any **three** questions from each section in separate script.

Section – A

1. (a) How will you define Urban Development Plan? Arrange different forms of urban development plan practiced in Bangladesh in a hierarchical order. (07)
(b) Narrate the contribution of following tools in urban development plan implementation. Use appropriate example. (10)
(i) Regulation (ii) Fiscal instrument (iii) Public and private investment
(c) "Urban planning decisions are based on certain theories, standards and practices"-how? Explain with proper example. (10)
(d) "Urban development plan deals with diverse components of urban area"-provide a brief overview of those components. (08)
2. (a) "Master plan ensures comprehensive development including future layout of a city"-justify the statement with an example. (12)
(b) Identify some drawbacks of Master Plan regarding its implementation in Bangladesh. (05)
(c) "Strategic decisions and policy formulation are the logical outcome of national development goals in Strategic Plan"-explain with an example. (10)
(d) Make a comparative analysis between the two forms of Local Plan- Action Area Plan and Subject Plan. Which form is mostly practiced in Bangladesh? (08)
3. (a) Make a comparison between the operational procedure of Guided Land Development and Land Readjustment Plan techniques of residential area development. Which one is more feasible regarding socio-economic condition of Bangladesh? (15)
(b) "Land Banking technique is a vital policy instrument of government for controlling spontaneous development and price of land"- explain. Use necessary example. (13)
(c) "Land Readjustment Plan is not applicable in already developed area"-why? Explain. (07)
4. (a) Narrate the directions of Dhaka Metropolitan Building Construction Rules, 2007 regarding following issues: (15)
(i) Minimum setback from boundary (ii) Necessary arrangement of roads
(iii) Corner simplification
(b) Write short notes on the following issues (any four): (20)
(i) Master Plan preparation process (ii) Format of Structure Plan
(iii) Fixing priority of projects in Detailed Area Plan
(iv) Land ceiling technique for ensuring equitable distribution of resources
(v) Access facilities for fire service

Section – B

5. (a) Make a list of data and information needed for urban planning. (05)
- (b) Why information on population is an important part of planning information? (06)
Mention some measurable criteria to have an idea of any country's population.
- (c) Why do planners prioritize population forecast over estimate and projection? (07)
Population extrapolation is a two stage process- explain.
- (d) Below figures are population projection (pyramid) of Bangladesh and Japan. (17)
What is your understanding about the future economic effects of the population composition of the two countries?



6. (a) Define basic and non-basic industry. Explain the standard industrial (08)
classification system.
- (b) What is Income Multiplier (IM) and what are the assumptions and weakness of (07)
the IM?
- (c) Why employment is an important factor for an economy? Project the economy (20)
for 2015 from the following data set using shift-share approach.

Industry \ Year	Local Employment (in millions)		Observed Economy in a Ref. Region (in millions)	
	1998	2005	1998	2003
Construction (SIC-C)	1529	1378	227.2	179.2

7. (a) Why is housing study important for Urban and Regional Planners? (10)
 - (b) What is the difference between need and demand? Is housing a need or (08)
demand? –explain.
 - (c) Show a conceptual framework for housing choice analysis. Having enough (08)
supply of housing will not solve housing problem-explain.
 - (d) Make a list of data requirement when planning for housing development. (09)
8. (a) What is built-out analysis? Explain the mapping and interpretation technique of (12)
the built-out analysis.
 - (b) What is land suitability analysis? What are the steps to land suitability (11)
analysis? Explain the steps in short.
 - (c) Mention major criteria for residential/commercial/industrial area's land (08)
suitability analysis.
 - (d) What are the applications of land suitability analysis? (04)